

LANGZHONG, China

WMO: 573060

Lat: 31.5846N

Lon: 105.9776E

Elev: 1263

StdP: 14.04

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 34.0	(c) 35.9	(d) 18.2	(e) 14.4	(f) 47.0	(g) 24.4	(h) 19.4	(i) 47.3	(j) 9.7	(k) 44.4	(l) 8.2	(m) 45.0	(n) 1.8	(o) 320	(p) 0.311

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 13.4	(c) 96.4	(d) 77.8	(e) 93.9	(f) 77.3	(g) 91.5	(h) 76.6	(i) 80.9	(j) 90.7	(k) 79.8	(l) 89.0	(m) 78.8	(n) 87.2	(o) 3.9	(p) 140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.4	154.4	85.6	77.4	149.5	84.5	76.5	145.0	83.6	45.9	91.0	44.7	89.0	43.5	87.2	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.6	(b) 7.7	(c) 6.8	(e) 30.7	(f) 99.5	(g) 2.3	(h) 2.5	(i) 29.0	(j) 101.3	(k) 27.7	(l) 102.7	(m) 26.4	(n) 104.1	(o) 24.8	(p) 105.9		
(a) 9.6	(b) 7.7	(c) 6.8	(e) 28.6	(f) 82.2	(g) 2.5	(h) 1.6	(i) 26.8	(j) 83.4	(k) 25.4	(l) 84.3	(m) 24.0	(n) 85.2	(o) 22.2	(p) 86.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	64.1	44.0	48.4	56.3	65.5	72.2	77.3	82.0	81.9	73.4	64.7	55.7	46.5	(6)
(7)		DBStd	14.02	3.92	4.85	6.19	6.28	4.76	4.31	4.25	5.31	5.45	5.03	5.09	3.99	(7)
(8)		HDD50	413	188	80	13	1	0	0	0	0	0	0	11	120	(8)
(9)		HDD65	2388	650	466	277	68	6	0	0	3	67	279	572		(9)
(10)		CDD50	5550	3	34	207	466	687	819	993	987	702	456	183	13	(10)
(11)		CDD65	2049	0	0	7	83	228	369	528	522	254	57	1	0	(11)
(12)		CDH74	18498	0	0	49	576	1573	3072	5772	5708	1567	179	2	0	(12)
(13)		CDH80	7556	0	0	5	147	464	1123	2576	2691	525	25	0	0	(13)
(14)	Wind	WSAvg	2.4	2.1	2.2	2.6	2.7	2.8	2.5	2.6	2.5	2.4	2.1	2.0	2.1	(14)
(15)	Precipitation	PrecAvg	40.9	0.5	0.6	1.3	2.7	4.6	4.2	8.0	6.9	7.2	3.1	1.3	0.5	(15)
(16)		PrecMax	66.9	1.1	1.7	3.1	8.5	14.2	9.3	18.1	23.6	21.1	8.3	2.8	1.2	(16)
(17)		PrecMin	24.5	0.0	0.1	0.2	0.6	0.7	0.5	1.3	0.9	0.2	0.9	0.1	0.0	(17)
(18)		PrecStd	10.0	0.3	0.4	0.7	1.6	3.1	2.1	3.9	4.5	4.9	1.6	0.7	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.6	67.5	79.3	88.8	91.6	94.2	98.8	99.8	95.4	83.3	72.5	60.5	(19)
(20)			MCWB	48.5	53.7	62.0	67.7	69.6	74.9	78.2	77.5	76.3	68.4	60.5	50.3	(20)
(21)		2%	DB	55.2	62.3	74.2	84.1	88.1	91.3	95.9	97.2	90.1	78.8	69.0	56.6	(21)
(22)			MCWB	46.5	50.8	59.9	65.8	68.8	74.7	78.3	77.9	75.0	66.8	59.3	48.8	(22)
(23)		5%	DB	52.3	59.0	70.6	80.4	85.0	88.9	93.4	94.7	85.9	75.3	65.5	54.3	(23)
(24)			MCWB	45.1	49.9	57.8	64.7	68.5	74.0	78.0	77.8	72.8	65.3	57.8	47.9	(24)
(25)		10%	DB	50.2	56.1	66.9	76.7	82.2	86.5	91.1	92.0	82.3	72.3	62.8	52.3	(25)
(26)			MCWB	44.4	48.5	56.1	63.5	67.8	73.4	77.6	77.3	71.8	64.4	57.2	47.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.0	55.2	64.8	71.2	74.7	79.7	82.7	82.0	79.1	71.2	62.8	53.5	(27)
(28)			MCDWB	55.7	64.1	76.3	84.1	84.4	88.6	92.4	93.4	89.7	79.2	68.5	56.6	(28)
(29)		2%	WB	47.9	52.8	61.2	68.7	72.7	77.9	81.1	80.8	77.0	68.8	61.2	51.3	(29)
(30)			MCDWB	52.9	59.2	71.4	78.6	81.5	85.6	90.4	91.3	85.9	74.9	65.8	54.5	(30)
(31)		5%	WB	46.5	51.1	59.0	66.7	71.3	76.5	80.1	79.8	75.2	67.4	59.9	49.7	(31)
(32)			MCDWB	50.7	56.6	68.1	76.3	79.8	84.0	88.9	89.5	82.6	72.7	63.8	52.7	(32)
(33)		10%	WB	45.3	49.6	57.2	64.8	69.7	75.4	79.1	78.7	73.5	65.7	58.5	48.4	(33)
(34)			MCDWB	49.2	54.9	65.0	74.1	78.2	82.8	87.2	87.8	79.5	70.6	61.7	51.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.6	10.8	13.7	15.3	14.9	13.2	13.4	14.4	11.2	10.1	9.7	8.6	(35)
(36)			MCDBR	14.4	16.8	20.2	22.1	21.0	18.7	17.9	19.0	17.9	16.0	15.1	12.4	(36)
(37)			MCWBR	7.5	8.2	8.2	7.7	6.4	5.5	4.8	4.5	5.2	5.9	6.9	6.7	(37)
(38)		5% WB	MCDWB	12.3	14.4	18.4	18.5	16.3	14.5	14.7	16.3	14.0	12.9	11.7	9.5	(38)
(39)			MCWBR	6.9	7.6	7.9	7.4	6.9	5.2	5.3	5.3	5.3	5.5	6.0	5.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.715	0.764	0.774	0.743	0.693	0.657	0.665	0.695	0.709	0.752	0.758	0.718	(40)	
(41)		taud	1.425	1.368	1.385	1.475	1.611	1.709	1.715	1.638	1.584	1.468	1.429	1.453	(41)	
(42)		Ebn at Noon	157	167	181	197	210	216	214	205	193	167	147	147	(42)	
(43)		Edh at Noon	83	96	101	95	84	75	75	80	81	86	82	77	(43)	
(44)	All-Sky Solar Radiation	RadAvg	562	668	983	1281	1378	1348	1499	1422	985	746	587	512	(44)	
(45)		RadStd	71	120	116	161	159	162	247	215	135	126	83	80	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.52	+1.15	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		+0.50	N/A	-1.53	N/A	N/A	N/A	-65	-142	+175	+68

Nomenclature: See separate page